Faculty of Computing

Artificial Intelligence and Machine Learning

IT2011

Year 2 Semester 1 (2025)

Global Student Migration & Higher Education Trends

Group: 2025-Y2-S1-MLB-B4G1-01

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1. Introduction and Problem Statement

The global landscape of higher education is increasingly influenced by student migration, driven by factors such as scholarship opportunities, visa policies, and academic performance. This project aims to predict the placement status (employed or unemployed) of international students' post-graduation using machine learning techniques. The problem stems from the need to understand how educational and socio-economic variables impact employability, providing insights for students, universities, and policymakers. With the rise in international student mobility, accurately forecasting placement outcomes can help tailor support systems and address disparities in job market access. Our study focuses on leveraging a comprehensive dataset to build predictive models, addressing a critical gap in actionable data-driven insights for global education trends.

2. Dataset Description

Source and Structure

Link: https://www.kaggle.com/datasets/atharvasoundankar/global-student-migration-and-higher-education-trends

We used the *Global Student Migration & Higher Education Trends* dataset from Kaggle, created by Atharva Soundankar. The dataset is provided as a single CSV file.
It contains data on international student migration flows and higher-education metrics across countries and years. Typical features include:

- Origin or Origin _country: the country from which students originate
- Destination_country: the country where students go for higher education
- Number_of_students (or similar): count of students migrating from origin to destination in that year
- Possibly additional features/columns related to higher-education trends (e.g., tertiary enrolment, outbound/inbound ratios, etc.)

Key Characteristics

- **Number of records**: - 5,000 observations (rows)
- **Number of features (columns)**: - 20
- **Feature types** - Categorical (e.g., origin country, host country), numerical (student counts, enrolment numbers).
- **Temporal span** - Data covers multiple years around 2019-2023.
- **Geographic span**: Global (various countries of origin and destination, enabling cross-country comparisons and migration flow analysis).
- **Data quality**: Does not contain missing values, Does not contain duplicate rows.
- **Distribution & balance**: The target variable is balanced, with approximately 50% of students marked as "Placed" and 50% as "Not Placed".

Why This Dataset

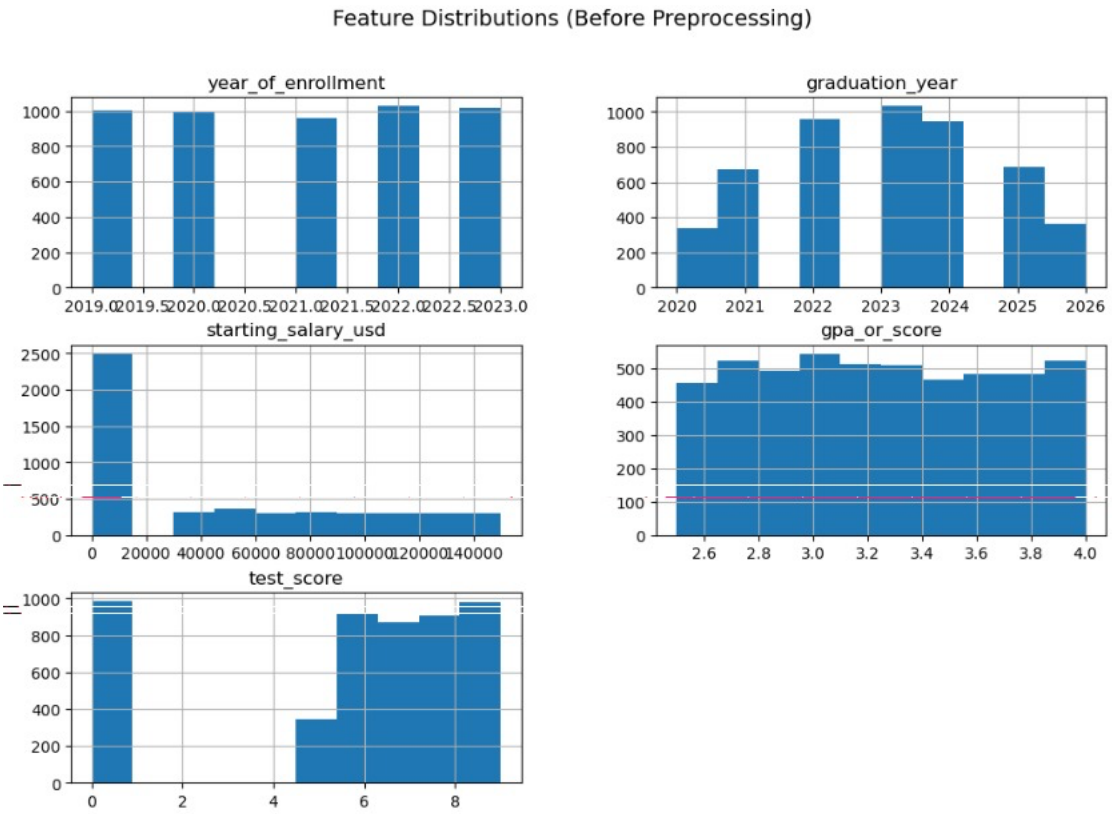
The dataset is well-suited for our project because:

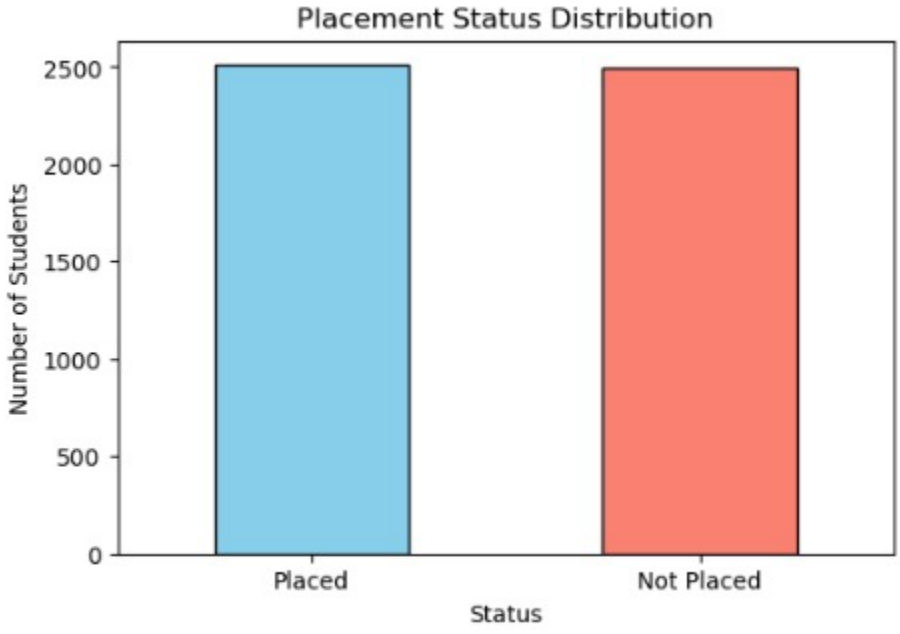
The dataset is ideal for our project because it is a real, labeled tabular dataset representing authentic student information and placement outcomes. The sufficient size (5,000 records) meets project requirements while remaining computationally manageable. The balanced target variable (50% Placed, 50% Not Placed) allows for fair model training and accurate performance evaluation.

3. Preprocessing & Exploratory Data Analysis

Our six-step preprocessing pipeline transforms raw data into a clean, machine-readable format:

1. **Scaling:** Standardizes numerical features to ensure equal contribution across variables.
2. **Label Encoding:** Converts categorical target labels into numeric form for model compatibility.
3. **One-Hot Encoding:** Transforms categorical features into binary columns to represent each category distinctly.
4. **Outlier Removal:** Detects and removes extreme values to improve model stability and accuracy.
5. **Feature Engineering:** Creates new informative features to enhance predictive performance.
6. **Dimensionality Reduction (PCA):** Reduces feature space while preserving maximum variance for efficient computation.


Feature Distributions (Before Preprocessing)
year_of_enrollment
graduation_year
starting_salary_usd
gpa_or_score
test_score


Placement Status Distribution
Number of Students
Placed
Not Placed
Status

4. Model Design and Implementation

We use five machine learning models were implemented and evaluated to predict the target outcome using the preprocessed dataset. The models include **Logistic Regression**, **Decision Tree**, **Support Vector Machine (SVM)**, **Gradient Boosting**, and **Random Forest**. Each model was tuned using both **GridSearchCV** and **RandomizedSearchCV** to identify the optimal hyperparameters.

Model	Avg. Accuracy	Avg. Precision	Avg. Recall	Avg. F1 Score	Avg. AUC	Best Tuning Method
SVM	0.524	0.527	0.489	0.507	0.522	GridSearchCV
Decision Tree	0.519	0.511	0.912	0.656	0.522	GridSearchCV
Gradient Boosting	0.503	0.523	0.578	0.550	0.519	RandomizedSearchCV
Logistic Regression	0.510	0.511	0.514	0.512	0.505	Both Equal
Random Forest	0.496	0.497	0.467	0.481	0.502	GridSearchCV

5. Evaluation and Comparison

We evaluated models using Accuracy, Precision, Recall, F1 Score, and AUC Score.

- **Logistic Regression** - Both tuning methods produced identical results, indicating that the logistic regression model is stable but not highly predictive for this dataset. Accuracy and AUC values near 0.5 suggest limited model generalization.
- **Decision Tree** - The Decision Tree showed high recall, meaning it correctly identified most positive cases, but precision and overall accuracy remained moderate. GridSearchCV slightly outperformed RandomizedSearchCV.
- **Gradient Boosting** - Both GridSearchCV and RandomizedSearchCV results show similar moderate performance. RandomizedSearchCV slightly improved recall and F1 Score, suggesting it captured more true positives.
- **Random Forest** - The Random Forest model achieved moderate accuracy with balanced precision and recall under GridSearchCV. The RandomizedSearchCV results were slightly lower, possibly due to suboptimal parameter sampling.
- **Support Vector Machine (SVM)** - The SVM model showed the best accuracy among all models with GridSearchCV (0.5236). Although still moderate, it performed better than logistic regression and random forest, suggesting it fits the data slightly more effectively.

Conclusion:
Among all models, **SVM using GridSearchCV** achieved the highest overall accuracy (0.5236) and AUC (0.5221). The Decision Tree had the best recall and F1 score, showing strength in detecting positive cases. However, all models performed moderately, indicating that further feature engineering or more advanced ensemble methods may be needed to improve accuracy.

6. Ethical Considerations and Bias Mitigation

Although our dataset is balanced, the correlation between features and the target variable is relatively low, which indicates weak predictive relationships. During preprocessing, we identified data leakage that could unfairly inflate model performance, so we removed specific columns to mitigate this issue. Despite addressing the data leakage, the gap between feature correlations and predictive performance remained, resulting in lower values across our evaluation metrics.

```
placement_status                                    1.000000
field_of_study_Engineering                          0.045214
destination_city_Vancouver                          0.035881
destination_city_Birmingham                         0.035158
destination_country_India                           0.031925
university_name_Simon Fraser University             0.027628
university_name_Higher School of Economics          0.027493
destination_city_Pilani                             0.026606
language_proficiency_test_IELTS                     0.025285
destination_city_Moscow                             0.024658
destination_city_Galway                             0.024257
destination_city_Kingston                           0.024164
university_name_University of British Columbia      0.023073
destination_city_Mumbai                             0.023073
destination_city_Manchester                         0.022019
university_name_Middlesex University Dubai          0.020875
destination_city_Chennai                            0.020505
enrollment_reason_Job Opportunities                 0.020382
destination_city_Tiruchirappalli                    0.020076
destination_country_United Kingdom                  0.019990
course_name_Data Science                            0.019132
course_name_Business Administration                 0.019009
university_name_University of Turku                 0.018926
enrollment_reason_Quality of Life                   0.018228
course_name_Biotechnology                           0.017924
university_name_Bauman Moscow State Technical University    0.017653
university_name_Lomonosov Moscow State University   0.017274
test_score                                          0.016011
destination_city_Bangalore                          0.015878
course_name_Economics                               0.015847
Name: placement_status, dtype: float64
```

7. Reflections and Lessons Learned

Throughout this project, we realized that having a balanced dataset alone does not guarantee strong model performance, especially when feature correlations with the target are low. The identification and correction of data leakage taught us the importance of careful preprocessing to ensure ethical and reliable results. We also learned that even after mitigating leakage, evaluation metrics can remain low if the underlying data does not strongly predict the outcome, highlighting the need for feature engineering and deeper data understanding. Overall, this experience reinforced the value of data integrity, critical analysis, and iterative improvement in building trustworthy machine learning models.

8. References

https://chatgpt.com/g/g-p-68ef7b9565848191ba1651ed069d53a8-ai-evaluation-2/project